1. B 2. A 3. B 4. A 5. C 6. D 7. D 8. C 9. A 10. B

11. D 12. D 13. C 14. C 15. A

16. ACD 17. ABD 18. BC 19. BD 20. AC 21. ABD 22. CD

23. AC 24. BD 25. BCD

Question 3

26.

(1) [9, 8, 6, 5, 4, 3, 2, 1]

(2) The average time complexity is O(len(lst)^len(lst))

27、

#1. len(lst)

#2. lst_len -1

#3. lst[i] > lst[i+1]

#4. temp = lst[i]

#5. lst[i] = lst[i+1]

#6. lst[i+1] = temp

#7. lst_len -= 1

28、

(1)
```
def fib(n)
    if n == 0 or n==1
        return 1
    elif n >= 2:
        return fib(n-1)+fib(n-2)
```

(2)
```
def fib(n):
    num1 = 1
    num2 = 1
    num_n = 0
    if n ==0 or n==1:
        return 1
    elif n>=2:
        for i in range(n-1):
            num_n = num1 + num2
            num1 = num2
            num2 = num_n
        return num_n
fib(n)
```

29.

(1) This code is aim to create a binary tree and do a depth-first search to a binary tree.

(2) Assume N as the number of edges in the tree, then the average time complecity is O(N)

(3) The output is:

2
5
9
4
7
6
11
5
2